



Class #8 - Matrix Exponential



Handout Class #8 Printout

1. If the characteristic polynomial of A does not have n distinct roots, then A may not have n linearly independent eigenvectors in $\mathbb{C}$.

Example:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

has only two distinct eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$.

alg. mult = 2

$$\text{Nul}(A - 1 \cdot I) = \text{Nul} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$p(\lambda)$ Degree n

Exactly n complex roots
repeated. $\nearrow$ real.

rank = 2
nullity = 1 = geo. mult.

Consequently, the DE $\dot{x} = Ax$ has only 2 linearly independent solutions $e^{\lambda_1 t} \vec{v}_1$ and $e^{\lambda_2 t} \vec{v}_2$.

$$e^{1 \cdot t} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$e^{2t} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

prob.

2. Problem: Suppose $n \times n$ matrix A has only $k < n$ linearly independent eigenvectors. Then, the DE $\dot{x} = Ax$ has only k linearly independent solutions of the form $e^{\lambda t} \vec{v}$.
How to find n linearly independent solutions?

3. Recall: The general solution to $\frac{dy}{dt} = ay$ is $y(t) = C e^{at}$.
Analogously, we would like to say that $x(t) = e^{At} v$ is a solution of the vector DE $\frac{dx}{dt} = Ax$ for every constant vector v . How do we define e^{At} though?

$$\frac{dy}{dt} = y' = a(y(t))$$

$$y(t) = C e^{at}$$

$$y' = C \cdot a e^{at} = a C e^{at} = a y$$

Def: For an $n \times n$ matrix A , the matrix exponential, denoted by e^{At} is defined as

Exam
DE
PP

$$e^{At} = I + At + \frac{(At)^2}{2!} + \dots + \frac{(At)^n}{n!} + \dots$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots$$

5. It is known in Linear Algebra that the series converges for all t , and can be differentiated term-by-term:

$$\frac{d}{dt} e^{At} = A + A^2 t + \dots = A \left(I + At + \frac{A^2 t^2}{2!} + \dots \right) = A e^{At}$$

This implies that $e^{At} v$ is a solution of (*) for every constant vector v .

$$\frac{d}{dt} \begin{bmatrix} e^{At} v \end{bmatrix} = A \begin{bmatrix} e^{At} v \end{bmatrix} = A e^{At} v$$

PP Examples.

$$10 \mid 5$$

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...

6. Remark: The matrix exponential e^{At} and the scalar exponential e^{at} satisfy many similar properties, for example,

However,

$$(e^{At})^{-1} = e^{-At} \quad \text{and} \quad e^{A(t+s)} = e^{At} e^{As}.$$

e^{At} is invertible.

$$e^{At+Bt} = e^{At} e^{Bt} \iff AB = BA \quad (A \text{ and } B \text{ commute})$$

(Proofs are done in Linear Algebra.)

7. Back to #2:

Observe that

for any scalar λ since

Moreover, $e^{At}v = e^{\lambda t} e^{(A-\lambda I)t}v$.

$$v \in \text{Nul}(A - \lambda I)^m$$

Suppose v satisfies $(A - \lambda I)^m v = 0$ for some integer m , then the infinite series $e^{(A-\lambda I)t}v$ terminates after m terms, and

$$e^{At}v = e^{\lambda t} \left(v + t(A-\lambda I)v + \dots + \frac{(A-\lambda I)^{m-1} t^{m-1}}{(m-1)!} v \right)$$

geo. $\angle$ alg. with $L.I.$

$$\frac{d}{dt} \vec{x}(t) = A \vec{x}(t)$$

Physics

$$\dot{x} = \frac{d}{dt} x(t)$$

8. Algorithm for finding n linearly independent solutions of $\dot{x} = Ax$.

(i) Find all eigenvalues and eigenvectors of A . If A has n linearly independent eigenvectors:

$$e^{At} v^i$$

is a soln.

$$i=1, \dots, n \quad e^{(A-\lambda I)t} v^i = e^{\lambda t} (I - (A-\lambda I)t + \frac{(A-\lambda I)^2 t^2}{2!} - \dots) v^i$$

$\{e^{At} v^1, \dots, e^{At} v^n\} \in \text{Nul}(A-\lambda I)$ L.I. eigenvectors.

$(A-\lambda I)v^i = 0$
 $A v^i = \lambda v^i$

(The infinite series $e^{(A-\lambda I)t} v$ terminates after first term.)

(ii) Suppose A has only $k < n$ linearly independent eigenvectors (so, have k linearly independent solutions). To find additional solutions we pick an eigenvalue λ of A and find all vectors v such that $(A - \lambda I)^2 v = 0$, but $(A - \lambda I)v \neq 0$. For each such vector v ,

$$e^{At} v = e^{\lambda t} e^{(A-\lambda I)t} v = e^{\lambda t} (v + (A-\lambda I)t v)$$

is an additional solution.

So that

3) alg. mult. > 1
 2) geo. mult. > 1

$$\text{Nul}(A-\lambda I)^2 \supseteq \text{Nul}(A-\lambda I)$$

$$\text{Nul}(A-\lambda I)$$

only one L.I.

$$(A-\lambda I)x=0 \Rightarrow (A-\lambda I)^2 x=0$$

$$\text{Basis of } \text{Nul}(A-\lambda I) = \{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \}$$

$$\text{Basis} = \{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \}$$

(iii) If we still do not have enough solutions, then we find all vectors v for which $(A-\lambda I)^3 v = 0$, but $(A-\lambda I)^2 v \neq 0$. For each such vector v ,

$$(A-\lambda I)^2 v \neq 0$$

$$e^{At} v = e^{\lambda t} (v + t(A-\lambda I)v + \frac{t^2}{2!} (A-\lambda I)^2 v)$$

is an additional solution.

$$\text{Nul}(A-\lambda I)^3 \supseteq \text{Nul}(A-\lambda I)^2 \supseteq \text{Nul}(A-\lambda I)$$

(iv) We keep proceeding in this manner until we obtain n linearly independent solutions.

1.9 per.

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